Asset	Security Requirement No.	Security Requirement Category	Security Requirement	Security Requirement Description	Threat Scenario No.
E1.1. GPS/IMU	SR-1	AI/Sensor Robustness & Input Validation	Signal plausibility checks and multi-sensor fusion algorithms shall be implemented to verify GNSS signal validity.	The system must employ algorithms that cross-validate GNSS inputs against other vehicle sensors (e.g., odometer, IMU, LiDAR) and analyze signal characteristics (SNR, power levels) to filter out spoofed signals.	T1, T2, T3, T5
	SR-2	Availability & Resilience	A fail-safe mechanism shall be triggered automatically upon detection of GNSS jamming or signal loss.	In the event of high-power noise injection or signal disruption, the system must immediately switch to a degraded mode (e.g., Dead Reckoning) to maintain minimal safe operation capabilities.	T6
	SR-3	Communication Security & Integrity	Internal communication messages containing GNSS data shall be protected with Message Authentication Codes (MAC) and freshness counters.	To prevent replay and injection attacks on the CAN/Ethernet bus, all GNSS-related frames must include a cryptographic tag for integrity and a monotonically increasing counter for freshness.	T7, T8
	SR-4	Authentication & Access Control	Access to GNSS configuration interfaces and middleware topics shall be restricted to authenticated entities only.	Interfaces capable of altering sensor settings (e.g., OBD-II, debug ports) and DDS topics must enforce strict access control lists (ACLs) to block unauthorized reconfiguration or data publishing.	T4, T9
	SR-5	Security Auditing & Monitoring	Security anomalies and signal discrepancy events shall be recorded in a tamper-proof secure storage.	All detected instances of spoofing, jamming, or authentication failures must be securely logged with synchronized timestamps to facilitate post-incident forensics and non-repudiation.	T10
E1.2. Radar	SR-6	AI/Sensor Robustness & Input Validation	Interference mitigation and signal modulation techniques shall be implemented to validate radar signal authenticity.	The system must employ techniques such as randomizing chirp parameters (frequency hoping, jittering) and consistency checks (Doppler/Range correlation) to detect and reject spoofed signals or AoA manipulation.	T11, T12, T13, T14, T15
	SR-7	Communication Security & Integrity	Internal communication messages containing radar object lists shall be protected with Message Authentication Codes (MAC) and freshness counters.	To mitigate CAN injection and replay attacks, all frames transmitting radar-detected object data (Range, Azimuth) must carry cryptographic integrity tags and freshness values.	T16, T17
	SR-8	Authentication & Access Control	Physical sensor connections and middleware communication topics shall be restricted to authenticated and authorized entities.	A hardware-based device authentication mechanism must be enforced to prevent rogue sensor connection, and DDS middleware must utilize strict Access Control Policies to block unauthorized publishers.	T18, T19
	SR-9	Security Auditing & Monitoring	Detected radar signal anomalies and unauthorized access attempts shall be recorded in a tamper-proof secure storage.	Security events, including signal spoofing detection (e.g., mismatch in chirp parameters), jamming detection, and authentication failures, must be securely logged for incident investigation.	T20
E1.3. LiDAR	SR-10	AI/Sensor Robustness & Input Validation	LiDAR signal validity checks and multi-modal sensor fusion shall be implemented to detect spoofing and interference.	The system must implement algorithms to analyze pulse characteristics (e.g., pulse width, return intensity) and cross-validate point clouds with Camera/Radar data to reject fake echoes and excessive noise.	T21, T22, T23
	SR-11	Communication Security & Integrity	Internal LiDAR data transmission shall be protected using Message Authentication Codes (MAC) and freshness verification mechanisms.	To prevent packet injection and replay attacks on the internal network (Ethernet/CAN), point cloud data frames must be encapsulated with cryptographic integrity tags and synchronized timestamps.	T24, T25
	SR-12	Authentication & Access Control	Mutual authentication and strict access control policies shall be enforced for LiDAR hardware connections and middleware communication topics.	The system must require cryptographic device authentication (e.g., 802.1X for Ethernet) for physical ports and enforce DDS permissions to prevent rogue publishers from injecting fake data.	T26, T27
	SR-13	Security Auditing & Monitoring	Security anomalies involving LiDAR signal manipulation or unauthorized access shall be recorded in tamper-proof logs.	Events such as scan matching failures, unexpected point clusters (ghost objects), or authentication errors must be securely logged to enable post-incident forensics and attack attribution.	T28
E1.4. Ultrasonic	SR-14	AI/Sensor Robustness & Input Validation	Signal modulation and pulse validity checks shall be implemented to distinguish genuine echoes from spoofing signals.	To mitigate ToF spoofing (distance manipulation) and crosstalk/jamming, the sensor system must employ pulse coding (e.g., specific frequency patterns or chirps) and validate the pulse width/timing of received echoes against the transmitted pattern.	T29, T30
	SR-15	Communication Security & Integrity	Ultrasonic data frames transmitted over the vehicle network shall be protected with Message Authentication Codes (MAC) and freshness counters.	To prevent the injection of fake distance data (e.g., "30cm" stop signal) and replay of stale "Clear Path" messages, all sensor frames on CAN/Ethernet must include cryptographic integrity tags and sequence counters.	T31, T32
	SR-16	Authentication & Access Control	Authentication and strict access controls shall be enforced for sensor configuration APIs, hardware connections, and middleware topics.	Access to calibration APIs and DDS topics must be restricted to authorized processes to prevent sensitivity tampering or ghost data injection. Additionally, hardware authentication should be used to detect unauthorized sensor replacements.	T33, T34, T35
	SR-17	Security Auditing & Monitoring	Raw signal anomalies and security-critical events related to ultrasonic sensors shall be recorded in tamper-proof logs.	The system must securely log instances of abnormal signal patterns (potential spoofing), jamming detection, and authentication failures to distinguish between sensor faults and malicious attacks during post-incident analysis.	T36
E1.5. Camera	SR-18	AI/Sensor Robustness & Input Validation	Adversarial defense mechanisms and optical anomaly detection algorithms shall be implemented to validate visual inputs.	The system must incorporate defenses against physical adversarial patches (e.g., printed stickers), phantom attacks (e.g., projections targeting rolling shutter), and sensor blinding (e.g., laser saturation) by verifying temporal consistency and using multi-modal sensor fusion.	T37, T38, T39, T40
	SR-19	Communication Security & Integrity	Video stream data transmitted over internal networks shall be protected using cryptographic integrity checks and freshness mechanisms.	To prevent the injection of manipulated frames (e.g., traffic light color change) and replay of stale video feeds, high-throughput video streams must be authenticated (e.g., partitioned MAC) and timestamped to ensure data freshness.	T41, T42, T44
	SR-20	Authentication & Access Control	Mutual authentication shall be enforced for physical camera interfaces to prevent unauthorized device connections.	A secure handshake protocol (e.g., MIPI CSI-2 security extension or challenge-response) must be implemented to verify that only authorized camera modules are connected to the ECU, blocking rogue simulators.	T43
	SR-21	Security Auditing & Monitoring	Detected adversarial attempts, sensor saturation events, and classification anomalies shall be securely logged for forensic analysis.	Security-relevant events, including detection of adversarial patterns, blooming effects, or unexpected drops in classification confidence, must be recorded in tamper-proof logs to aid in accident investigation and liability clarification.	T45
E2. Vehicle Gateway & Actuators	SR-22	Communication Security & Integrity	Control commands and status messages transmitted across the vehicle network shall be protected using Message Authentication Codes (MAC) and freshness counters.	To prevent masquerading, command injection, and replay attacks on critical control buses (CAN/Ethernet), all control frames (e.g., Brake/Steer) and status feedbacks must be authenticated and checked for freshness.	T46, T50, T51
	SR-23	Authentication & Access Control	The Gateway shall implement strict firewall policies and routing filtering rules to isolate critical control networks from external interfaces.	The Gateway must enforce a whitelist-based filtering policy to block unauthorized messages originating from external interfaces (e.g., OBD-II, Telematics) from crossing into safety-critical domains (e.g., Chassis, Powertrain).	T47, T49, T52
	SR-24	System Hardening & Platform Security	Cryptographic keys and administrative credentials shall be stored in a secure hardware root of trust and shall not be hardcoded in the firmware.	To prevent credential theft via firmware analysis, all sensitive secrets (e.g., SSH keys, Gateway routing certificates) must be securely managed using HSM (Hardware Security Module) or secure storage mechanisms.	T48
	SR-25	Authentication & Access Control	Diagnostic access (e.g., UDS services) and configuration interfaces shall require cryptographic authentication prior to granting execution privileges.	Access to privileged diagnostic services (e.g., UDS Security Access) and Gateway configuration ports must be guarded by strong authentication (e.g., Seed & Key, Certificate exchange) to prevent unauthorized control or rule modification.	T47, T52
	SR-26	Security Auditing & Monitoring	Abnormal control traffic and unauthorized access attempts shall be recorded in a tamper-proof secure log.	The system must log security events such as blocked packets by the firewall, authentication failures on diagnostic ports, and replay detection events to facilitate forensic analysis and attribution.	T53
E3. Driver	SR-27	AI/Sensor Robustness & Input Validation	Voice recognition systems shall implement liveness detection and spectral analysis algorithms to reject replay and synthesized speech attacks.	To mitigate audio replay and AI-generated voice cloning attacks, the ASV system must verify the physical presence of the speaker using challenge-response or acoustic feature analysis (e.g., detecting electronic noise or lack of breathing sounds).	T54, T55
	SR-28	Authentication & Access Control	Multi-factor authentication (MFA) and secure distance bounding protocols shall be enforced for remote control and passive keyless entry.	To prevent relay attacks (T56) and session hijacking (T57, T58), the system must utilize UWB-based Time-of-Flight (ToF) checks for physical keys and strong cryptographic session management for remote apps.	T56, T57, T58
	SR-29	Authentication & Access Control	Input validation and Role-Based Access Control (RBAC) shall be applied to all Human-Machine Interface (HMI) and control command paths.	To prevent privilege escalation via infotainment (Wi-Fi/BT) and injection of fake driver commands, strict RBAC must be enforced, and all internal command messages must be verified for origin authenticity.	T59, T60, T61

	SR-30	Security Auditing & Monitoring	Critical driver interactions, including handover approvals and mode changes, shall be logged securely to ensure non-repudiation.	All safety-critical driver inputs, such as approving a transition to autonomous mode, must be logged with a digital signature and precise timestamp to serve as legal evidence in case of disputes.	T62
P1. Device Driver	SR-31	System Hardening & Platform Security	Strict input validation and memory safety mechanisms shall be implemented for all IOCTL and data processing routines.	To prevent kernel memory corruption (e.g., buffer overflows, integer overflows, use-after-free), all data received from user space or hardware must undergo rigorous bounds checking, type validation, and race condition (TOCTOU) protection.	T68, T69, T70, T73, T80, T81, T83
	SR-32	Authentication & Access Control	Source authentication and data integrity verification shall be enforced prior to initializing devices or processing sensor streams.	Drivers must implement handshake protocols to verify the authenticity of connected hardware (preventing rogue devices) and verify digital signatures/checksums on data packets to reject malformed or spoofed inputs.	T63, T64, T65, T66, T67
	SR-33	System Hardening & Platform Security	Secure boot and system hardening measures shall be applied to protect driver firmware and configuration files.	To prevent persistent compromise and privilege escalation, the system must enforce Secure Boot (preventing unsigned firmware/rootkits), disable debug interfaces/backdoors in production, and set restrictive file permissions on configuration files.	T71, T72, T74, T75
	SR-34	Availability & Resilience	Resource rate limiting and side-channel defenses shall be implemented to ensure system availability and confidentiality.	The driver must implement interrupt throttling to prevent livelock (DoS) from bus flooding and utilize compiler mitigations (e.g., Retpolines) to protect against speculative execution attacks (Spectre) leaking kernel memory.	T78, T79, T82, T84
	SR-35	Security Auditing & Monitoring	Security-critical driver operations and exceptions shall be audited with caller identification details.	To trace malicious activities, the driver must log critical events such as IOCTL calls with anomalous parameters, DDS security exceptions, and unauthorized access attempts, including the Process ID (PID) of the caller.	T76, T77
P2.1. OS	SR-36	System Hardening & Platform Security	Kernel interfaces and system services shall validate all inputs and enforce memory safety protections to prevent code execution.	To mitigate buffer overflows, integer errors, and insecure deserialization, the OS must employ exploit mitigations (ASLR, DEP/NX, Stack Canaries) and strictly sanitize inputs from IOCTLs, IPC, and serialized data streams.	T90, T91, T93, T94, T101, T106, T107
	SR-37	System Hardening & Platform Security	A hardware-rooted Secure Boot chain shall be implemented to verify the integrity of the bootloader and kernel.	To prevent persistent compromises via BootROM exploits (RCM), bootkit installation, or modification of startup scripts, the system must verify the digital signature of every boot stage starting from an immutable hardware root of trust.	T92, T103, T105, T108
	SR-38	Authentication & Access Control	Middleware and network services shall enforce mutual authentication and bind to specific internal interfaces only.	To prevent unauthorized node connections and injection attacks, ROS/DDS middleware must utilize security plugins (e.g., DDS Security) for authentication, and IPC sockets/services must not bind to wildcard addresses (0.0.0.0).	T85, T86, T89
	SR-39	Availability & Resilience	Resource management policies and interrupt rate limiting shall be implemented to prevent Denial of Service.	To protect real-time tasks from network flooding or SMMU error storms, the OS must enforce resource quotas, CPU affinity, and interrupt throttling mechanisms to isolate faults and maintain availability.	T102, T104
	SR-40	System Hardening & Platform Security	Sensitive memory regions and execution contexts shall be isolated using hardware-enforced protection domains.	To mitigate side-channel attacks (Spectre), KASLR bypass, and SMM extraction, critical secrets (keys) and privileged code (SMM/TrustZone) must be isolated from user space using memory protection units and side-channel resistant coding practices.	T98, T99, T100
	SR-41	Data Confidentiality & Privacy	Data at rest and in transit shall be protected using strong encryption and authenticity verification.	All stored sensitive files must be encrypted, and network packets must carry MACs to prevent tampering. Session keys must be managed securely to prevent hijacking.	T87, T88, T97
	SR-42	Security Auditing & Monitoring	Security-relevant OS events and middleware transactions shall be logged to a centralized audit subsystem.	To enable detection of reconnaissance and intrusion, the OS must audit permission denials, DDS security events, and configuration changes, ensuring logs are protected from modification by attackers.	T95, T96
P2.2.1. Sensor Fusion	SR-43	Communication Security & Integrity	Input sensor data streams and middleware topics shall be authenticated and verified for integrity.	To prevent the ingestion of forged frames (ghost objects) from compromised nodes or the network, all incoming data from sensors and middleware (ROS/DDS) must be verified using Message Authentication Codes (MAC) and source authentication mechanisms.	T109, T110, T121, T124
	SR-44	AI/Sensor Robustness & Input Validation	Multi-modal consistency checks and outlier detection mechanisms shall be implemented to reject spoofed or anomalous sensor data.	The fusion algorithm must implement logic to cross-validate data between different modalities (e.g., LiDAR vs. Camera) and filter out physically impossible inputs (outliers) to prevent model corruption and resource exhaustion attacks.	T111, T112, T123
	SR-45	AI/Sensor Robustness & Input Validation	Strict input validation and safe deserialization routines shall be enforced to prevent code injection and memory corruption.	To mitigate vulnerabilities such as buffer overflows, path traversal, and insecure deserialization, all inputs including configuration files, serialized maps, and IPC messages must undergo rigorous validation and sanitization.	T113, T114, T115, T116, T117
	SR-46	Supply Chain & Lifecycle Security	The integrity of AI models, maps, and executable files shall be verified using cryptographic signatures before loading.	To prevent the execution of backdoored models or compromised fusion binaries (Supply Chain Attacks), the system must verify the digital signature of all static assets and executables against a trusted root certificate upon loading or updating.	T125, T126
	SR-47	System Hardening & Platform Security	Fusion processes and sensitive state data shall be isolated using hardware-enforced memory protection or Trusted Execution Environments (TEE).	To defend against memory leaks, side-channel attacks, and isolation bypasses, critical fusion logic and data must be executed within protected memory regions or TEEs (e.g., TrustZone), preventing unauthorized access from lower-privilege processes.	T120, T122, T127
	SR-48	Security Auditing & Monitoring	Discrepancies between sensor inputs and unauthorized access attempts shall be recorded in a tamper-proof audit log.	Security-relevant events, including sensor data mismatch alerts (indicating potential spoofing), validation failures, and unauthorized service access, must be securely logged to enable post-incident analysis and improvements in sensor logic.	T118, T119
P2.2.2. Localization	SR-49	Communication Security & Integrity	Localization data streams and middleware topics shall be authenticated to prevent the ingestion of forged PVT or map data.	To mitigate the injection of fake Position/Velocity/Time (PVT) frames or spoofed ROS topics, the system must enforce message authentication (MAC) and publisher validation on all network interfaces.	T128, T130
	SR-50	Supply Chain & Lifecycle Security	HD Maps and configuration files shall be cryptographically signed and verified integrity before loading.	To prevent map poisoning (T132, T146) and TOCTOU attacks (T136), High-Definition (HD) maps and config files must be digitally signed by a trusted authority and verified by the bootloader or application at runtime.	T132, T136, T145, T146
	SR-51	AI/Sensor Robustness & Input Validation	Strict input sanitization and secure deserialization mechanisms shall be implemented to prevent code execution and memory corruption.	The system must validate all inputs against strict schemas to prevent command injection, integer overflows, and insecure deserialization of map objects, ensuring memory safety.	T133, T134, T135, T143, T147
	SR-52	Availability & Resilience	Robust scan matching algorithms and resource limits shall be applied to maintain availability against spoofing and DoS attacks.	To defend against LiDAR/Radar noise injection (T131) and resource exhaustion (T142), localization algorithms (e.g., ICP/NDT) must include outlier rejection logic and enforce CPU/Memory quotas.	T129, T131, T142, T144
	SR-53	Data Confidentiality & Privacy	Location history logs and internal state data shall be encrypted at rest and in transit.	To protect vehicle trajectory privacy and sensitive state information from leakage via file system access or IPC sniffing, all location logs and inter-process communications must be encrypted.	T139, T140, T141
	SR-54	Security Auditing & Monitoring	Localization failures, spoofing detection events, and security configuration changes shall be logged securely.	Security-critical events such as map matching divergences (potential spoofing), authentication failures, and changes to DDS permissions must be recorded in tamper-proof logs for auditability.	T137, T138
P2.2.3. InterPretation & Prediction	SR-55	AI/Sensor Robustness & Input Validation	Adversarial defense mechanisms and out-of-distribution (OOD) detection shall be implemented to validate input data quality.	To mitigate physical adversarial attacks (patches, stickers) and sensor spoofing (phantom objects), the system must employ defenses such as adversarial training, input sanitization, and multi-modal consistency checks to reject malicious inputs.	T151, T152, T544, T545, T546, T547, T548, T549, T550, T556
	SR-56	Communication Security & Integrity	Middleware topics and IPC channels used for interpretation data shall be authenticated to prevent data injection.	To block the injection of fake object tracks or manipulated attributes (e.g., velocity), the prediction module must verify the digital signature and source authenticity of all incoming data frames from the middleware layer.	T148, T149, T150
	SR-57	Supply Chain & Lifecycle Security	AI models and configuration files shall be cryptographically signed and validated prior to loading to ensure supply chain integrity.	To prevent the deployment of poisoned models (backdoors), unauthorized weight modifications, or deserialization exploits, all model artifacts must be signed by a trusted authority and strictly validated (safe deserialization) before use.	T153, T155, T164, T165, T551, T554, T555
	SR-58	Availability & Resilience	Input size limits and computational resource quotas shall be enforced to prevent Denial of Service (DoS) and Sponge attacks.	The system must implement strict bounds on input dimensions and monitor execution time to prevent resource exhaustion caused by sponge examples, memory floods, or integer overflow triggers.	T154, T162, T163, T552
	SR-59	Data Confidentiality & Privacy	Sensitive operational data and model parameters shall be encrypted and protected against inference and side-channel attacks.	To protect PII (trajectory), proprietary model IP (weights), and prevent model inversion/extraction, data must be encrypted, and inference APIs must employ rate limiting and differential privacy techniques.	T158, T159, T160, T161, T553, T557
	SR-60	Security Auditing & Monitoring	Prediction decisions, confidence scores, and security anomalies shall be securely logged for explainability and non-repudiation.	The system must log inference metadata (model version, decision threshold) and security events (authentication failures) in a tamper-proof manner to support accident investigation and liability attribution.	T156, T157

P2.3. Planning	SR-61	Communication Security & Integrity	Middleware communication channels for planning inputs shall be authenticated to prevent unauthorized trajectory injection.	To mitigate the injection of fake trajectories or manipulated status frames (e.g., emergency stop), the planning module must verify the digital signature and source authenticity of all incoming ROS/DDS messages.	T167, T168
	SR-62	AI/Sensor Robustness & Input Validation	Adversarial defense mechanisms and input sanitization shall be implemented to detect semantic perturbations and prompt injections.	The system must employ defenses against physical adversarial attacks (stickers, patches), digital noise in cost maps, and indirect prompt injections (e.g., malicious text on signs) to ensure robust decision-making.	T169, T172, T560, T561, T565, T567, T569
	SR-63	Supply Chain & Lifecycle Security	Cryptographic integrity verification shall be enforced for HD Maps, AI models, and training datasets.	To prevent data poisoning (label flipping), backdoor insertion, and map modification, all static assets used for planning must be digitally signed and verified against a trusted root before usage.	T558, T562, T568, T571
	SR-64	System Hardening & Platform Security	Secure boot and strict input validation shall be implemented to prevent arbitrary code execution and memory corruption.	The system must validate deserialization processes and input parameters to prevent RCE, memory overflows, and privilege escalation. Additionally, Secure Boot must protect the boot chain (CBoot/MB2) from tampering.	T170, T171, T173, T180, T181, T182, T183
	SR-65	Availability & Resilience	Computational resource quotas and input complexity limits shall be enforced to prevent Denial of Service.	To defend against "Sponge Examples" (computationally expensive queries) and ghost object flooding, the planner must enforce strict time and memory budgets for optimization tasks and reject overly complex inputs.	T179, T564, T570
	SR-66	Data Confidentiality & Privacy	Sensitive planning data and model parameters shall be encrypted and protected against extraction attacks.	To prevent the theft of proprietary planning policies (IP) and user privacy leakage, data must be encrypted at rest/transit, and API access must be monitored to detect model extraction or inversion attempts.	T176, T177, T178, T559, T563, T566, T572
	SR-67	Security Auditing & Monitoring	Planning logic metadata and security events shall be logged in a tamper-proof manner for auditability.	The system must securely log cost map weights, decision constraints, and security violations (e.g., unauthorized permission changes) to explain controversial decisions and attribute liability in case of accidents.	T174, T175
P2.4. Mode Manager	SR-68	Communication Security & Integrity	Strict authentication and freshness verification mechanisms shall be enforced for all mode transition requests and handover status messages.	To prevent unauthorized engagement of autonomous modes or forced handovers via spoofing/replay (e.g., fake "System Ready"), the Mode Manager must verify the digital signature and timestamp/nonce of all incoming status messages from HMI and middleware.	T184, T185, T186, T187
	SR-69	Supply Chain & Lifecycle Security	Configuration files and policy documents shall be cryptographically signed, and all input parameters shall undergo strict sanitization and safe deserialization.	To mitigate RCE and logic corruption caused by tampering with handover policies (xml) or deserialization attacks, all configuration assets must be signed, and inputs via IPC must be sanitized against injection attacks.	T188, T189, T190, T191, T200, T201, T202
	SR-70	Availability & Resilience	Resource rate limiting and message queue quotas shall be implemented to maintain responsiveness during high-load or flooding events.	To ensure the system can process a "Handover Request" even under DoS conditions (e.g., status message flooding), the Mode Manager must enforce CPU/Memory quotas and prioritize critical control messages over diagnostic traffic.	T197, T198, T199
	SR-71	Data Confidentiality & Privacy	Driver authority state and handover history data shall be encrypted at rest and in transit.	To prevent the leakage of sensitive operational data (who was driving) and driver behavior patterns, all authority status logs and internal notification channels must be encrypted using strong cryptographic algorithms.	T195, T196
	SR-72	Security Auditing & Monitoring	Mode transitions, handover events, and security exceptions shall be logged with precise, synchronized timestamps in a tamper-proof manner.	To enable liability clarification (driver vs. system) after an accident, the exact timing of handovers, unauthorized mode change attempts, and authentication failures must be securely logged to ensure non-repudiation.	T192, T193, T194
P2.5. Control	SR-73	Communication Security & Integrity	Control command messages and middleware topics shall be authenticated and protected against replay attacks using SecOC mechanisms.	To prevent the execution of fake path tracking commands, replay of stale acceleration frames, and spoofing of planning nodes, the control module must enforce strict message authentication (MAC) and freshness checks (Monotonic Counters) on all incoming control topics.	T203, T204, T205, T206, T207, T220
	SR-74	AI/Sensor Robustness & Input Validation	Strict input sanitization and memory safety checks shall be enforced for control parameters and state objects.	To mitigate RCE and system crashes caused by parameter tampering (e.g., PID gains), unsafe deserialization, or integer overflows, all configuration inputs and serialized state objects must undergo rigorous validation before processing.	T208, T209, T210, T211, T223
	SR-75	Authentication & Access Control	Access to control configuration files, binaries, and driver interfaces shall be restricted to authorized processes.	To prevent privilege escalation and unauthorized hardware register modification, strict file permissions (removing world-writable flags) and kernel-level access controls must be applied to control binaries and IOCTL interfaces.	T212, T222, T221
	SR-76	Availability & Resilience	Real-time resource constraints and message rate limiting shall be implemented to ensure control loop stability.	To defend against DoS attacks targeting the control loop (e.g., topic flooding), the system must enforce rate limiting on input messages and guarantee CPU time slots for critical control tasks to prevent actuator stalls.	T219
	SR-77	Data Confidentiality & Privacy	Proprietary control parameters and tuning data shall be encrypted during storage and transmission.	To protect intellectual property (IP) such as PID gains and vehicle dynamics models from theft, all tuning parameters stored in logs or transmitted over the network (calibration) must be encrypted.	T216, T217, T218
	SR-78	Security Auditing & Monitoring	Control decisions, error conditions, and security violations shall be logged with tamper-proof mechanisms.	To facilitate post-incident analysis and distinguish between network faults and attacks, the system must securely log active control modes, thresholds, authentication failures, and timeout events.	T213, T214, T215
P2.6.1. Infotainment	SR-79	AI/Sensor Robustness & Input Validation	Voice recognition interfaces shall implement liveness detection to reject synthetic (Deepfake) and replayed audio.	To prevent unauthorized command execution via Deepfake voice clones or recorded audio, the ASR (Automatic Speech Recognition) system must verify the physical presence of the speaker using spectral analysis or challenge-response mechanisms.	T224
	SR-80	AI/Sensor Robustness & Input Validation	All user inputs, including text, voice, and files, shall be sanitized to prevent Command Injection, Prompt Injection, and Path Traversal.	The system must enforce strict schema validation and sanitization on all inputs to block OS command injection (shell metacharacters), LLM prompt injection (adversarial instructions), and path traversal attempts in file uploads.	T229, T230, T231, T231(DS-64)
	SR-81	Authentication & Access Control	Internal communication and sessions shall be authenticated and protected against spoofing and hijacking.	To prevent Source ID spoofing (e.g., faking Mode Manager) and session hijacking, all HMI communications must use authenticated endpoint binding, Message Authentication Codes (MAC), and secure session tokens with expiration enforcement.	T225, T226, T227, T228
	SR-82	Supply Chain & Lifecycle Security	Software updates, plugins, and serialized objects shall be cryptographically verified prior to execution.	To mitigate code execution via malicious plugins or insecure deserialization, the system must verify the digital signature of all external resources and software updates, and utilize safe deserialization methods.	T232, T242, T243
	SR-83	Data Confidentiality & Privacy	Sensitive user data (voice, location, PII) shall be encrypted at rest and in transit, and debug interfaces shall be disabled.	To protect user privacy from log leakage, sniffing, or memory dumps, voice recordings and location history must be encrypted, and debug ports (JTAG/UART) must be physically or logically disabled in production.	T236, T237, T238
	SR-84	Availability & Resilience	Input rate limiting and resource management shall be applied to prevent Availability attacks.	To defend against audio/text flooding or alert storms that freeze the screen, the system must enforce input rate limits and prioritize critical safety alerts over general infotainment traffic.	T239, T240
	SR-85	Security Auditing & Monitoring	Security-critical user interactions and errors shall be logged to a secure audit trail.	To ensure auditability, the system must log authentication outcomes, ASR execution statuses, and error events, while ensuring that the logs themselves do not contain plaintext sensitive user data.	T233, T234, T235
	SR-86	AI/Sensor Robustness & Input Validation	Multi-modal input sanitization and alignment mechanisms shall be implemented to detect and reject jailbreaks and adversarial injections.	To defend against direct/indirect prompt injection (e.g., "Ignore rules"), visual adversarial patches, and hidden text in images, the system must employ robust input filtering, adversarial training, and safety alignment (e.g., RLHF) to block malicious instructions.	T247, T249, T534, T543, T576, T577, T578, T581, T589
	SR-87	System Hardening & Platform Security	The MLLM shall operate within a strictly sandboxed environment with least privilege to prevent unauthorized code execution and excessive agency.	To mitigate risks from malicious code generation (e.g., Python scripts) and unauthorized actuator calls, the model must run in a sandbox without direct OS shell access, and critical actions must require "Human-in-the-loop" confirmation.	T250, T538, T583, T592
	SR-88	AI/Sensor Robustness & Input Validation	Context integrity verification and RAG security measures shall be enforced to prevent hallucination and knowledge poisoning.	To prevent responses based on poisoned RAG sources, conflicting knowledge, or prompt worms, the system must validate the relevance and source reputation of retrieved context and detect self-replicating patterns in the session history.	T535, T540, T575, T579, T582, T595

P2.6.2. MLLM	SR-89	Supply Chain & Lifecycle Security	Supply chain security controls including model signing and backdoor scanning shall be applied to all AI artifacts.	To prevent the deployment of poisoned models (sleeper agents), malicious fine-tuning, or compromised datasets, all models and weights must be cryptographically signed, and deserialization processes must be secured against RCE payloads.	T248, T259, T260, T537, T541, T573, T574, T584, T593
	SR-90	Data Confidentiality & Privacy	Defenses against model extraction, inversion, and system prompt leakage shall be implemented to protect intellectual property and privacy.	To protect the system prompt, proprietary knowledge, and PII from leakage via embedding inversion or proxy model training, the system must implement output filtering, differential privacy, and limit API query granularity.	T253, T254, T255, T536, T539, T580, T585, T586, T587, T588, T594
	SR-91	Availability & Resilience	Resource quotas and algorithmic complexity limits shall be enforced to prevent Denial of Service via sponge attacks.	To defend against algorithmic DoS attacks (e.g., sponge examples, token bombs) that exhaust GPU memory or energy, the system must enforce strict limits on input token length, recursion depth, and inference time per request.	T256, T257, T542, T590
	SR-92	Security Auditing & Monitoring	Inputs, outputs, and safety filter decisions involving MLLM interactions shall be logged securely for accountability.	To determine liability in case of AI-induced accidents, the system must securely log the raw input prompt, the generated response, and the logic of safety guardrails (e.g., why a command was allowed/blocked).	T251, T252
D1.1. Sensor Log & Evidence	SR-93	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization mechanisms shall be implemented for the log collection pipeline.	To prevent Remote Code Execution (RCE) via malicious serialized objects and arbitrary file overwrites via Path Traversal, the logging system must strictly validate input types, verify file paths, and use safe deserialization libraries.	T261, T264
	SR-94	Authentication & Access Control	Strict Access Control Lists (ACLs) and cryptographic integrity protection shall be enforced for log files and storage directories.	To prevent unauthorized deletion or tampering of evidence logs, file system permissions must be restricted to privileged processes, and integrity mechanisms (e.g., digital signatures, HMAC, or Chained Hashes) must be applied to detect modification.	T262, T263
	SR-95	Availability & Resilience	Storage quotas, log rotation policies, and rate limiting shall be implemented to ensure service availability.	To defend against Denial of Service (DoS) attacks causing disk exhaustion or I/O saturation, the system must enforce storage limits, rotate logs automatically, and rate-limit incoming log events to prevent evidence loss during floods.	T268, T269
	SR-96	Data Confidentiality & Privacy	Sensitive data within logs shall be masked, and log storage shall be encrypted at rest and in transit.	To prevent the leakage of PII (location history) and authentication tokens, logs must undergo data masking (anonymization) before writing, and all storage media and transmission channels must be encrypted.	T265, T266
	SR-97	System Hardening & Platform Security	Debug interfaces shall be disabled, and memory buffers containing log data shall be sanitized after use.	To prevent memory scraping attacks that extract residual log data or layout information, debug ports (JTAG/UART) must be disabled in production, and memory regions must be zeroed out after processing sensitive logs.	T267
D1.2. Planning Log & Evidence	SR-98	Authentication & Access Control	Strict Access Control Lists (ACLs) and cryptographic integrity verification shall be enforced for planning logs.	To prevent unauthorized deletion and tampering of decision logs (e.g., trajectory coordinates), file access must be restricted to privileged processes, and integrity must be protected using digital signatures or hash chains.	T271, T272
	SR-99	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE and path traversal attacks.	To mitigate code execution via malicious serialized log objects and file overwrites via manipulated filenames, the system must use safe deserialization libraries and strictly sanitize all file paths.	T270, T273
	SR-100	Availability & Resilience	Log rate limiting and storage quotas shall be enforced to prevent Denial of Service (DoS) and evidence loss.	To defend against disk exhaustion caused by error log loops or bus flooding, the logging service must limit the rate of incoming messages and reserve storage space for critical decision evidence.	T277, T278
	SR-101	Data Confidentiality & Privacy	Planning logs shall be encrypted at rest and in transit, and debug information shall be masked in production.	To protect driver privacy (routes, future intentions), logs containing trajectory data must be encrypted, and debug modes that expose detailed intent must be disabled or masked in production environments.	T274, T275
	SR-102	System Hardening & Platform Security	Memory buffers containing planning data shall be sanitized, and physical debug interfaces shall be disabled.	To prevent the extraction of sensitive operational data (recent decisions, keys) via cold boot attacks or JTAG, volatile memory must be zeroed out after use, and physical debug ports must be locked.	T276
D1.3. HMI Log & Evidence	SR-103	Authentication & Access Control	Access control and cryptographic integrity protection shall be enforced for HMI logs to prevent tampering.	To ensure forensic reliability and prevent liability shifting (e.g., changing "Accelerate" to "Brake"), log files must be protected by strict ACLs and digital signatures/HMACs to detect any unauthorized modification or deletion.	T279, T280, T281
	SR-104	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent code execution and model poisoning.	To mitigate RCE via serialized log objects and prevent the injection of adversarial prompts into future training datasets (RLHF poisoning), the logging pipeline must validate inputs and filter/tag malicious content before storage.	T282, T283
	SR-105	Availability & Resilience	Storage quotas and input rate limiting shall be enforced to ensure log availability against flooding attacks.	To defend against DoS attacks where attackers flood voice/touch events to exhaust disk space or I/O queues, the system must enforce strict storage limits per user/session and rate-limit logging requests.	T287, T288
	SR-106	Data Confidentiality & Privacy	Sensitive PII and conversational data shall be encrypted at rest and in transit.	To protect user privacy (voice, contacts, navigation) from leakage via file access or network sniffing, all HMI logs must be encrypted, and highly sensitive fields should be masked or tokenized in general logs.	T284, T285
	SR-107	System Hardening & Platform Security	Volatile memory buffers containing user interactions shall be sanitized, and debug interfaces disabled.	To prevent the reconstruction of chat history or voice buffers from memory dumps, the system must zero out memory after processing and disable debug interfaces (e.g., core dumps) in production builds.	T286
D1.4. Mode Manager Log	SR-108	Authentication & Access Control	Cryptographic integrity protection (e.g., Digital Signatures) and strict Access Control shall be enforced for Mode Manager logs.	To prevent liability shifting by tampering with "Authority Status" (e.g., changing Auto to Manual) or deleting accident logs, file access must be restricted to privileged processes, and integrity must be protected using digital signatures or hash chains.	T290, T291
	SR-109	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE and data poisoning.	To mitigate Remote Code Execution (RCE) via malicious serialized log objects and prevent the injection of false Takeover Request (TOR) data that could poison future model training, inputs must be strictly validated and sanitized.	T292, T293, T289
	SR-110	Availability & Resilience	Log rate limiting and storage reservation policies shall be enforced to ensure availability of handover evidence.	To defend against DoS attacks where attackers trigger a storm of mode transitions to exhaust disk space or I/O queues, the system must limit the rate of logging and reserve dedicated storage for critical handover events.	T297, T298
	SR-111	Data Confidentiality & Privacy	Sensitive operational data in logs shall be encrypted at rest and in transit, and debug modes disabled in production.	To protect driver privacy (ID, location, behavior) and operational status from leakage via network sniffing or memory dumps, logs must be encrypted, and debug information must be masked or disabled in production.	T294, T295, T296
D1.5. Control Log	SR-112	Authentication & Access Control	Cryptographic integrity mechanisms and strict Access Control Lists (ACLs) shall be enforced for control logs.	To prevent tampering with actuation data (e.g., faking brake application) or unauthorized deletion of accident evidence, control logs must be protected by digital signatures/HMACs and restricted to privileged access only.	T300, T301
	SR-113	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE and log destruction.	To mitigate RCE via malicious serialized log objects and prevent the destruction of actuation evidence via path traversal attacks, inputs must be strictly validated and sanitized before processing.	T299, T302
	SR-114	Availability & Resilience	Log rate limiting and storage quotas shall be enforced to ensure the availability of accident reconstruction data.	To defend against DoS attacks where attackers flood dummy control events to exhaust disk space or I/O queues, the system must limit logging rates and reserve storage for critical emergency maneuvers.	T306, T307
	SR-115	Data Confidentiality & Privacy	Proprietary control parameters in logs shall be encrypted or masked, and debug logging disabled in production.	To protect intellectual property (IP) such as PID gains and safety limits from theft via log extraction or sniffing, sensitive parameters must be encrypted or masked, and debug logging must be disabled in production builds.	T303, T304
	SR-116	System Hardening & Platform Security	Physical debug interfaces shall be disabled, and volatile memory containing control data shall be sanitized.	To prevent the retrieval of sensitive control strategies or cryptographic keys from memory artifacts via JTAG/UART, physical debug ports must be disabled, and memory buffers must be sanitized after use.	T305
DF1.1. Position, Motion Signal	SR-117	Communication Security & Integrity	Position and motion signal data flows shall be protected using Message Authentication Codes (MAC) and freshness values.	To prevent the injection of rogue NMEA frames, MITM modification, and replay of old PVT data, all position/motion signals transmitted over the bus must include cryptographic integrity tags (MAC) and monotonically increasing counters or timestamps.	T308, T309, T310, T311
	SR-118	Availability & Resilience	Bandwidth monitoring and rate limiting mechanisms shall be implemented to detect and mitigate flooding attacks on signal buses.	To defend against Denial of Service (DoS) caused by high-frequency garbage frames or replayed signals saturating the IRQ handler, the communication controller must enforce rate limits and detect traffic anomalies.	T315
	SR-119	Data Confidentiality & Privacy	Sensitive position and motion data transmitted over internal buses shall be encrypted to protect user privacy.	To prevent the leakage of driver location history and movement patterns via physical tapping or sniffing, data payloads containing precise coordinates or velocity history must be encrypted using strong cryptographic algorithms.	T312, T313, T314
	SR-120	AI/Sensor Robustness & Input Validation	Signal validity checks and timeout mechanisms shall be implemented to handle signal loss or interference gracefully.	To maintain system safety during jamming or interference events (T316), the receiver must implement timeout logic and signal quality checks (e.g., CRC, error flags) to detect data flow interruptions and trigger fail-safe modes.	T316
DF1.2. Radar	SR-121	Communication Security & Integrity	Radar data transmission shall be protected using Message Authentication Codes (MAC) and freshness counters.	To prevent the injection of fake object lists (masquerading), replay of old frames (phantom vehicles), and MITM modification, radar data frames on Ethernet/CAN-FD must include cryptographic integrity tags and freshness values.	T317, T318, T319, T321
	SR-122	AI/Sensor Robustness & Input Validation	Signal validity checks and interference detection mechanisms shall be implemented to handle RF spoofing and jamming.	To mitigate the effects of RF spoofing (ghost targets) and jamming (SNR drop), the data flow processing logic must validate physical plausibility (e.g., Doppler consistency) and monitor signal quality metrics to trigger fail-safe modes.	T320, T326

Signal	SR-123	Data Confidentiality & Privacy	Internal radar data streams containing object lists or raw maps shall be encrypted.	To prevent the leakage of real-time environmental perception (surrounding vehicles/pedestrians) and sensitive sensor parameters via sniffing, high-bandwidth radar data (e.g., LVDS, Ethernet) must be encrypted.	T322, T323, T324
	SR-124	Availability & Resilience	Input rate limiting and buffer management shall be enforced to prevent DoS via radar frame flooding.	To defend against denial of service attacks where the bus or IRQ handler is saturated with noise packets, the communication controller must enforce strict rate limits and drop invalid radar frames at the ingress point.	T325
DF1.3. Point Cloud Signal	SR-125	Communication Security & Integrity	Point Cloud data transmission shall be authenticated using efficient Message Authentication Codes (MAC) and freshness mechanisms.	To prevent the injection of rogue frames, replay of old scans, and MITM modification on high-bandwidth links (Ethernet), data must be authenticated using efficient schemes (e.g., chained MACs or hardware-accelerated integrity tags) with freshness validation.	T327, T328, T329, T330
	SR-126	AI/Sensor Robustness & Input Validation	Signal characteristic validation and saturation detection shall be implemented to identify physical spoofing and blinding attacks.	To mitigate laser pulse injection (Ghost Objects) and sensor blinding (saturation), the processing logic must validate signal characteristics (e.g., pulse width, return intensity distribution) and detect abnormal saturation levels to exclude fake points.	T331, T336
	SR-127	Data Confidentiality & Privacy	Internal Point Cloud data streams shall be encrypted to prevent environmental mapping leakage.	To prevent the reconstruction of high-precision 3D maps of the vehicle's surroundings via sniffing (Privacy/Surveillance), raw XYZ and intensity data transmitted over internal networks must be encrypted.	T332, T333, T334
	SR-128	Availability & Resilience	Input rate limiting and network bandwidth management shall be enforced to prevent Denial of Service via frame flooding.	To defend against DoS attacks where the ECU's ring buffer or network stack is saturated with flood packets, the system must enforce bandwidth quotas and rate limits on the LiDAR input interface.	T335
DF1.4. UltraSonic Range Signal	SR-129	Communication Security & Integrity	Ultrasonic signal transmission shall be protected using lightweight Message Authentication Codes (MAC) and freshness counters.	To prevent rogue device injection, replay of old distance frames, and MITM attacks on low-bandwidth buses (e.g., LIN), data must be authenticated using bandwidth-efficient integrity tags (e.g., checksums with keys) and counters.	T337, T339, T340
	SR-130	AI/Sensor Robustness & Input Validation	Pulse coding and signal saturation detection mechanisms shall be implemented to identify physical spoofing and jamming.	To mitigate false distance calculation (spoofing) and sensor blindness (jamming), the sensor processing logic must use coded pulses (chirps) to validate echoes and monitor noise levels to trigger safe degradation.	T338, T345
	SR-131	Data Confidentiality & Privacy	Sensitive proximity data and sensor activation patterns transmitted over internal links shall be encrypted.	To prevent the leakage of environmental geometry (e.g., garage layout) and driver behavior profiling via bus tapping, distance measurements and control signals must be encrypted to protect privacy.	T341, T342, T343
	SR-132	Availability & Resilience	Bus load monitoring and input rate limiting shall be enforced to prevent Denial of Service on low-bandwidth channels.	To defend against flooding attacks that saturate the LIN bus or interrupt handlers, the master controller must monitor bus load and enforce strict rate limits on slave responses.	T344
DF1.5. Image Signal	SR-133	Communication Security & Integrity	Image data transmission shall be protected using hardware-accelerated integrity checks and freshness validation.	To mitigate rogue device connection, replay of old video streams (empty road), and MITM pixel modification on high-speed links (CSI-2/GMSL), image frames must be authenticated using low-latency methods (e.g., CRC+Rolling Counter or Partial MAC) supported by hardware.	T346, T347, T348, T349
	SR-134	AI/Sensor Robustness & Input Validation	Optical anomaly detection algorithms shall be implemented to identify phantom projections and sensor saturation.	To defend against optical attacks such as phantom image projection (fake signs) and laser dazzling (sensor saturation), the perception pipeline must analyze temporal consistency, exposure levels, and noise patterns to reject invalid frames.	T350, T355
	SR-135	Data Confidentiality & Privacy	Video streams containing PII shall be encrypted or anonymized (blurred) in real-time.	To prevent the leakage of real-time surveillance video and privacy breaches (facial recognition, license plates) via sniffing, video streams must be encrypted, or PII regions must be blurred immediately upon capture if full encryption induces unacceptable latency.	T351, T352, T353
	SR-136	Availability & Resilience	Bandwidth monitoring and interface rate limiting shall be enforced to prevent video stream flooding.	To defend against DoS attacks where the video bus or ECU ring buffer is overwhelmed by garbage data or repeated frames, the deserializer/receiver interface must enforce strict bandwidth limits and drop excess packets.	T354
DF2. Sensor Raw Data	SR-137	Authentication & Access Control	Strict Access Control Lists (ACLs) and process isolation shall be enforced for inter-process communication (IPC) channels handling sensor raw data.	To prevent unauthorized injection (T356) or sniffing (T359) of raw sensor data within the OS, the kernel and middleware must restrict access to shared memory segments and ROS topics to authenticated producer/consumer processes only.	T356, T359
	SR-138	Communication Security & Integrity	Internal sensor data structures shall include integrity checksums and timestamp/sequence fields to detect memory corruption and replay.	To mitigate in-memory tampering (T357) and replay of stale buffers (T358), raw data frames passed between processes must be wrapped with internal validation headers (e.g., CRC, monotonic counters).	T357, T358
	SR-139	Availability & Resilience	Input validation and resource usage limits shall be applied to internal data processing pipelines.	To defend against local DoS attacks via flooding (T362) or malformed inputs (T363), the system must enforce message queue limits, validate header integrity before processing, and implement robust exception handling to prevent pipeline stalls.	T362, T363
	SR-140	System Hardening & Platform Security	Cryptographic keys used for internal protection shall be stored in secure hardware, and unnecessary sensitive fields shall be minimized.	To prevent data leakage via compromised keys (T360) or excessive logging (T361), keys must be managed via HSM/TEE (not plain config files), and high-precision location data should be minimized/stripped if not strictly required for the specific process.	T360, T361
DF2.1. Perception Data	SR-141	Communication Security & Integrity	Middleware communication shall enforce publisher authentication and data integrity verification.	To prevent masquerading (fake camera nodes) and tampering with object lists during transmission, the middleware (e.g., DDS/ROS2) must utilize security plugins to authenticate publishers and verify Message Authentication Codes (MAC) on data payloads.	T364, T365, T367, T368
	SR-142	Communication Security & Integrity	Data freshness checks shall be implemented to prevent the replay of stale perception frames.	To mitigate replay attacks where old perception streams are re-injected to confuse the fusion algorithm, data frames must include monotonically increasing sequence numbers or synchronized timestamps that are verified by the subscriber.	T366
	SR-143	Authentication & Access Control	Access Control Policies shall be enforced to restrict topic subscription and decryption.	To prevent unauthorized monitoring of object tracks (privacy) and decryption of internal states, strict Access Control Lists (ACLs) must be applied to middleware topics, allowing only authorized fusion nodes to subscribe and decrypt the data.	T369, T370
	SR-144	Availability & Resilience	Input validation and traffic shaping shall be implemented to defend against DoS and malformed data injection.	To protect the fusion service from crashing due to malformed inputs (e.g., NaN coordinates) or topic flooding, subscribers must validate data ranges/types and the middleware must enforce traffic shaping (QoS policies) to limit message rates.	T372, T373
	SR-145	Data Confidentiality & Privacy	Internal debug vectors and confidence scores shall be minimized or stripped from production data streams.	To prevent IP leakage and adversarial analysis (Model Inversion), detailed debug information and raw confidence scores that reveal model limitations should be removed from the production data stream or encrypted separately.	T371
DF2.2. a-priori Perception	SR-146	Communication Security & Integrity	Middleware handling map and pose data shall enforce strict authentication and integrity verification.	To prevent the injection of fake map tiles/GPS frames (T374), masquerading as a Map Server (T376), and MITM modification of map attributes (T377), the middleware (ROS2/DDS) must authenticate publishers and verify Message Authentication Codes (MAC) on all payloads.	T374, T376, T377
	SR-147	Communication Security & Integrity	Data freshness checks shall be implemented to prevent the replay of outdated map or location data.	To mitigate replay attacks where old map versions or past GPS coordinates are injected (T375), causing path deviation, all data frames must include synchronized timestamps or version numbers that are strictly validated by the consumer.	T375
	SR-148	Data Confidentiality & Privacy	Map data and location streams shall be encrypted to protect Intellectual Property (IP) and user privacy.	To prevent the theft of proprietary HD Map data and leakage of user location history via sniffing (T378) or compromised keys (T379), map assets and pose streams must be encrypted using secure key management practices (HSM).	T378, T379
	SR-149	Availability & Resilience	Input validation and resource quotas shall be enforced to defend against DoS via map flooding or malformed files.	To protect the localization service from stalling due to flood attacks (T381) or CPU exhaustion from parsing corrupted map files (T382), the system must validate file formats (e.g., OpenDRIVE XML) and enforce buffer/queue limits.	T381, T382
	SR-150	Data Confidentiality & Privacy	Metadata in location streams shall be minimized to prevent user profiling.	To prevent the inference of driver habits and life patterns from excessive metadata (POIs, quality metrics) (T380), data streams should be stripped of unnecessary context information before transmission or logging.	T380
DF2.3. Localization Data	SR-151	Authentication & Access Control	Middleware handling localization data shall enforce strict publisher authentication and topic access control.	To prevent unauthorized injection of false pose frames (T383) and masquerading as a Localization Node (T385), the middleware (e.g., ROS2/DDS) must enforce strict authentication policies (SROS) and allow only authorized nodes to publish to critical topics.	T383, T385
	SR-152	Communication Security & Integrity	Localization data integrity and freshness shall be protected using cryptographic signatures and timestamps.	To mitigate tampering with velocity/heading values and replay of old frames (T384), data messages must include Message Authentication Codes (MAC) and synchronized timestamps/counters to ensure integrity and freshness.	T384
	SR-153	Data Confidentiality & Privacy	Localization data streams containing precise pose and velocity shall be encrypted.	To prevent the leakage of real-time vehicle movement and user trajectory via IPC tapping or sniffing (T388, T389), payloads containing X, Y, Z coordinates and velocity vectors must be encrypted.	T388, T389
	SR-154	Availability & Resilience	Input validation and traffic shaping shall be implemented to prevent DoS via flooding or malformed data.	To protect downstream modules from crashing due to invalid covariance matrices (T392) or topic flooding (T391), subscribers must strictly validate schema/ranges (e.g., NaN checks) and middleware must enforce rate limits.	T391, T392
	SR-155	AI/Sensor Robustness & Input Validation	Cross-validation logic shall be implemented to reject propagated sensor spoofing attacks.	To prevent external GNSS spoofing from polluting the localization data flow (T387), the processing logic must cross-validate inputs (e.g., checking GNSS against IMU/Odometry) and reject physically impossible jumps in position.	T387
	SR-156	Data Confidentiality & Privacy	Unnecessary debugging metadata shall be stripped from localization streams to minimize privacy risks.	To prevent the inference of past/future routes from excessive debug info (Path IDs, Raw GNSS) (T390), data flows must be sanitized to include only the minimal information required for control.	T390
	SR-157	Authentication & Access Control	Middleware handling fusion data shall enforce strict publisher authentication and data integrity verification.	To prevent the injection of fake object tracks (T393), masquerading as a Fusion Node (T395), and modification of object attributes (T394), the middleware (ROS2/DDS) must enforce authentication policies and verify Message Authentication Codes (MAC) on all payloads.	T393, T394, T395
	SR-158	Communication Security & Integrity	Data freshness mechanisms shall be implemented to prevent the replay of stale environment models.	To mitigate the replay of old object tracks or environment states (T394) which could lead to collisions based on past reality, data frames must include synchronized timestamps or monotonic counters validated by the Planning module.	T394

DF2.4. Fusion Data	SR-159	AI/Sensor Robustness & Input Validation	Fusion logic shall implement anomaly detection and consistency checks to reject propagated sensor spoofing.	To prevent upstream sensor spoofing (e.g., Phantom Objects) from solidifying in the fusion output (T397), the fusion algorithm must implement robustness checks (e.g., temporal consistency, multi-modal validation) to filter out anomalies.	T397
	SR-160	Availability & Resilience	Input validation and traffic shaping shall be enforced to defend against DoS and malformed data injection.	To protect the Planning module from saturation via flooding (T401) or crashes due to invalid data types (T402), subscribers must validate inputs (schema/range) and the middleware must enforce rate limiting (QoS).	T401, T402
	SR-161	Data Confidentiality & Privacy	Fusion data streams containing object tracks and internal states shall be encrypted.	To prevent privacy breaches regarding surrounding traffic (T398) and theft of proprietary fusion logic via metadata analysis (T400), the environment model data flow must be encrypted.	T398, T400
	SR-162	Data Confidentiality & Privacy	Internal metadata and debug vectors shall be minimized or stripped to protect Intellectual Property.	To prevent attackers from reverse-engineering the fusion logic (e.g., confidence thresholds) (T400), unnecessary metadata and debug vectors must be removed from the production data stream.	T400
DF3.1. Sensor Log Data	SR-163	Authentication & Access Control	The logging service shall enforce strict access control and authentication to prevent unauthorized log injection.	To prevent low-privileged processes from injecting fake sensor logs (T403), the logging daemon or service must verify the identity and permissions of any process attempting to write to the log stream (e.g., via authenticated sockets or IPC).	T403
	SR-164	Communication Security & Integrity	Sensor log data in transit shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of log values (e.g., removing error codes), replay of old logs, and MITM attacks (T404, T405), log packets must include integrity tags (MAC) and timestamps/counters that are verified by the collector.	T404, T405
	SR-165	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization mechanisms shall be implemented to prevent code execution via malicious log objects.	To defend against Remote Code Execution (RCE) attacks where attackers inject malicious serialized objects into the log stream (T406), the collector must use safe deserialization libraries and strictly validate input types.	T406
	SR-166	Availability & Resilience	Rate limiting, traffic shaping, and storage quotas shall be implemented to ensure log availability under flooding conditions.	To protect the logging pipeline from being overwhelmed by noise logs or exception cascades (T409, T410), the system must enforce input rate limits, prioritize critical logs (QoS), and manage buffer/storage quotas.	T409, T410
	SR-167	Data Confidentiality & Privacy	Sensor log data streams shall be encrypted to protect sensitive environmental and location information.	To prevent the leakage of real-time location (GPS) and environmental context via tapping or sniffing (T407, T408), log streams carrying sensitive sensor metadata must be encrypted during transmission using strong cryptography.	T407, T408
DF3.2. Planning Log Data	SR-168	Authentication & Access Control	The logging interface shall enforce strict source authentication to prevent unauthorized decision log injection.	To prevent non-planning processes from injecting fake decision logs (e.g., falsely claiming "Obstacle Detected") (T411), the logging daemon must authenticate the identity of the process writing to the planning log stream via secure IPC or socket credentials.	T411
	SR-169	Communication Security & Integrity	Planning log transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of cost maps/path reasons and replay of old logs during transit (T412, T413), log payloads must include integrity tags (MAC) and synchronized timestamps that are verified by the collector before storage.	T412, T413
	SR-170	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE via log streams.	To defend against Remote Code Execution (RCE) attacks where attackers inject malicious serialized objects into the planning log stream (T414), the collector must use safe deserialization libraries and validate the structure of log records.	T414
	SR-171	Availability & Resilience	Rate limiting and prioritization (QoS) shall be enforced to ensure availability of critical decision logs.	To protect the logging pipeline from saturation due to debug floods or exception storms (T417, T418), the system must enforce rate limits and prioritize critical decision logs over verbose debug/error messages.	T417, T418
	SR-172	Data Confidentiality & Privacy	Planning log streams shall be encrypted to protect Intellectual Property (IP) and future trajectories.	To prevent the theft of proprietary algorithms (cost weights, decision trees) and leakage of planned paths via sniffing (T415, T416), log streams containing internal planning logic must be encrypted during transmission.	T415, T416
DF3.3. HMI Log Data	SR-173	Authentication & Access Control	The logging interface shall enforce strict source authentication to prevent fake HMI log injection.	To prevent non-HMI processes from injecting fake logs (e.g., falsely claiming "Driver ignored warning") (T419), the logging daemon must verify the identity of the process writing to the HMI log stream to prevent liability shifting.	T419
	SR-174	Communication Security & Integrity	HMI log transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of alert history or hands-on status (T420, T421) and replay attacks during transit, log payloads must include integrity tags (MAC) and timestamps that are verified by the collector before storage.	T420, T421
	SR-175	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE via log streams.	To defend against Remote Code Execution (RCE) attacks where attackers inject malicious serialized objects into the HMI log stream (T422), the collector must use safe deserialization libraries and strictly validate input structures.	T422
	SR-176	Availability & Resilience	Rate limiting and prioritization (QoS) shall be enforced to ensure availability of critical interaction logs.	To protect the logging pipeline from saturation due to touch event flooding or UI fault storms (T425, T426), the system must enforce rate limits and prioritize critical logs (e.g., Takeover attempts) over general events.	T425, T426
	SR-177	Data Confidentiality & Privacy	HMI log streams containing PII shall be encrypted to protect user privacy.	To prevent the leakage of sensitive personal data (voice transcripts, texts, biometrics) via sniffing or tapping (T423, T424), HMI log streams must be encrypted during transmission using strong cryptography.	T423, T424
DF3.4. Mode Manager Data	SR-178	Authentication & Access Control	The logging interface shall enforce strict source authentication to prevent fake mode log injection.	To prevent unauthorized processes from injecting fake logs (e.g., falsely claiming "Driver Requested Takeover") (T427) to shift liability, the logging daemon must verify the identity and privileges of the process writing to the mode log stream.	T427
	SR-179	Communication Security & Integrity	Mode Manager log transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of system failure codes (e.g., to "Driver Override"), timestamp alteration, and replay attacks (T428, T429), log payloads must include integrity tags (MAC) and timestamps that are verified by the collector.	T428, T429
	SR-180	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE via log streams.	To defend against Remote Code Execution (RCE) attacks where attackers inject malicious serialized objects into the mode log stream (T430), the collector must use safe deserialization libraries and strictly validate input structures.	T430
	SR-181	Availability & Resilience	Rate limiting and prioritization (QoS) shall be enforced to ensure availability of handover evidence.	To protect the logging pipeline from saturation due to mode oscillation floods or status update storms (T433, T434), the system must enforce rate limits and prioritize critical logs (e.g., Handover moments) over repetitive status updates.	T433, T434
	SR-182	Data Confidentiality & Privacy	Mode Manager log streams shall be encrypted to protect operational privacy.	To prevent the leakage of driver attention levels and handover habits via sniffing (T431, T432), log streams carrying operational mode details must be encrypted during transmission using strong cryptography.	T431, T432
DF3.5. Control Log Data	SR-183	Authentication & Access Control	The logging interface shall enforce strict source authentication to prevent fake control log injection.	To prevent non-critical processes from injecting fake control logs (e.g., falsely claiming "Brake: 100%") (T435) to contradict physical reality, the logging daemon must authenticate the identity of the process writing to the control log stream via secure IPC or socket credentials.	T435
	SR-184	Communication Security & Integrity	Control log transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of critical values (Throttle/Steering), removal of actuator errors, and replay attacks during transit (T436, T437), log payloads must include integrity tags (MAC) and timestamps that are verified by the collector before storage.	T436, T437
	SR-185	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE via log streams.	To defend against Remote Code Execution (RCE) attacks where attackers inject malicious serialized objects into the control log stream (T438), the collector must use safe deserialization libraries and strictly validate input structures.	T438
	SR-186	Availability & Resilience	Rate limiting and prioritization (QoS) shall be enforced to ensure availability of high-resolution accident logs.	To protect the logging pipeline from saturation due to dummy control updates or error storms (T441, T442), the system must enforce rate limits and prioritize high-resolution control logs required for accident reconstruction over general logs.	T441, T442
	SR-187	Data Confidentiality & Privacy	Control log streams containing proprietary tuning parameters shall be encrypted.	To prevent the theft of manufacturer's core control logic (IP) such as PID gains and torque limits via sniffing (T439, T440), log streams carrying these parameters must be encrypted during transmission using strong cryptography.	T439, T440
DF3.6. MLLM Log Data	SR-188	Authentication & Access Control	The logging interface shall enforce strict source authentication to prevent fake MLLM log injection.	To prevent unauthorized containers from injecting fake logs (e.g., falsely claiming "Safety Filter Triggered") (T443) to mask AI failures, the logging aggregator must verify the identity of the process/container writing to the MLLM log stream.	T443
	SR-189	Communication Security & Integrity	MLLM log transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate the modification of outputs (e.g., removing hate speech), alteration of System Prompt versions, and replay attacks (T444, T445), log payloads must include integrity tags (MAC) and timestamps verified by the collector.	T444, T445
	SR-190	AI/Sensor Robustness & Input Validation	Input sanitization and secure deserialization shall be implemented to prevent RCE via complex log objects.	To defend against RCE attacks via malicious serialized objects hidden in complex nested JSON/XML log entries (T446), the collector must use safe deserialization libraries and strictly validate input structures.	T446
	SR-191	Availability & Resilience	Log size limits, rate limiting, and prioritization (QoS) shall be enforced to ensure availability against token bombs.	To protect the logging pipeline from saturation due to massive prompt logs (Token Bomb) or crash loops (T449, T450), the system must enforce strict log size limits (truncation) and rate limiting.	T449, T450
	SR-192	Data Confidentiality & Privacy	MLLM log streams containing user PII and proprietary system prompts shall be encrypted.	To prevent the leakage of user voice commands (PII) and System Prompts (IP) via sniffing (T447) or decryption (T448), log streams must be encrypted using strong cryptography to protect both privacy and intellectual property.	T447, T448
DF4.1. Text/Audio Data	SR-193	Communication Security & Integrity	Input data streams shall be protected using strict source authentication, integrity checks, and freshness mechanisms.	To prevent unauthorized injection of commands (T451), replay of old voice commands (T452), and MITM modification (T453), the data flow must enforce source authentication, use Message Authentication Codes (MAC), and check timestamps/counters.	T451, T452, T453
	SR-194	AI/Sensor Robustness & Input Validation	Adversarial input detection and prompt sanitization mechanisms shall be implemented.	To defend against AI-specific attacks such as prompt injection (hidden instructions) and audio adversarial perturbations (T454), the system must implement input filters (e.g., perplexity checks, perturbation detection) before passing data to the MLLM.	T454
	SR-195	Availability & Resilience	Input size limits (Token Limits) and rate limiting shall be enforced to prevent Context Window Exhaustion (DoS).	To protect the MLLM from denial of service via huge prompts or continuous audio streams (T458), the system must enforce strict token count limits and rate limits per user session.	T458

	SR-196	Data Confidentiality & Privacy	Audio and text data streams containing user biometrics shall be encrypted.	To prevent the leakage of voiceprints (biometrics) and private conversations via wiretapping or sniffing (T455, T456), input streams must be encrypted from the capture point (Microphone/IVI) to the processing unit.	T455, T456
	SR-197	Data Confidentiality & Privacy	Metadata associated with user inputs shall be minimized to prevent privacy profiling.	To prevent the profiling of user emotions and habits (T457), excessive metadata (e.g., precise location, sentiment scores) should be stripped or anonymized if not strictly necessary for the command execution.	T457
DF4.2. Driver Command Code	SR-198	Authentication & Access Control	Middleware (DDS/IPC) shall enforce strict publisher authentication and Access Control Lists (ACLs).	To prevent unauthorized containers from injecting false commands (T459) and masquerading as legitimate nodes (T461), the middleware must verify the identity of the publisher using cryptographic certificates and enforce strict topic access policies.	T459, T461
	SR-199	Communication Security & Integrity	Command code transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate payload modification (e.g., Mode Change), replay of old "Unlock" commands (T460), and MITM attacks (T462), command packets must include integrity tags (MAC) and timestamps/counters verified by the control module.	T460, T462
	SR-200	AI/Sensor Robustness & Input Validation	A deterministic Safety Guardrail shall be implemented to validate MLLM-generated command codes before execution.	To prevent the propagation of malicious or unsafe commands generated by a compromised/hallucinating MLLM (T463), a rule-based validator must check the command against safety schemas and physical constraints (e.g., ignore unsafe "Go" commands).	T463
	SR-201	Availability & Resilience	Rate limiting and Quality of Service (QoS) policies shall be enforced to ensure command availability.	To protect the Planner from being overwhelmed by a flood of null commands (T467), the system must enforce rate limits per publisher and utilize QoS settings to prioritize time-critical commands (e.g., Brake).	T467
	SR-202	Data Confidentiality & Privacy	Command streams shall be encrypted, and debug flags/metadata stripped from production traffic.	To prevent the leakage of internal API structures (JSON schema), driver intent, and safety policies via sniffing (T464, T465, T466), command streams must be encrypted, and unnecessary debug flags must be removed.	T464, T465, T466
DF5. HMI Status & Alerts	SR-203	Authentication & Access Control	The HMI system shall enforce strict source authentication and process isolation for visual and audio alerts.	To prevent unauthorized apps from injecting false critical alerts (T468) or fake voice warnings (T473), the HMI must verify the source of every alert message and isolate audio channels to prevent authorized injections.	T468, T470, T473
	SR-204	Communication Security & Integrity	Status and alert messages shall be protected using Message Authentication Codes (MAC) and freshness checks.	To mitigate the suppression of collision warnings (T469) and modification of Autopilot status (Mode Confusion, T471), alert payloads must include integrity tags (MAC) and timestamps verified by the display unit.	T469, T471
	SR-205	Availability & Resilience	Alert prioritization (QoS) and rendering queue management shall be enforced to ensure display availability.	To defend against DoS attacks where the HMI is flooded with low-priority messages or error pop-ups (T477, T478), the system must prioritize critical safety alerts and limit the rate of non-critical notifications.	T477, T478
	SR-206	Authentication & Access Control	Session management for driver profiles shall be secured against hijacking and token theft.	To prevent the hijacking of user profiles and remote access via session token interception (T472), the login process must use secure key exchange protocols and protect session tokens in secure storage.	T472
	SR-207	Data Confidentiality & Privacy	Vehicle status streams shall be encrypted, and debug information stripped to protect privacy.	To prevent the leakage of real-time vehicle status (Speed, Location) and sensitive debug info (T474, T476), status streams must be encrypted, and unnecessary metadata must be removed from production traffic.	T474, T476
DF6. Control Data	SR-208	Authentication & Access Control	Middleware and actuators shall enforce strict source authentication to prevent unauthorized control command injection.	To prevent non-real-time processes or rogue nodes from injecting arbitrary torque/steering commands (T479, T481), the middleware (e.g., DDS/ROS2) and actuator controllers must authenticate the source of control messages.	T479, T481
	SR-209	Communication Security & Integrity	Control data transmission shall be protected using hardware-accelerated Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate suppression of "Brake" commands, replay of old acceleration commands, and MITM modification of steering angles (T480, T482), control packets must include MACs and freshness counters verified within real-time deadlines (e.g., via SecOC).	T480, T482
	SR-210	Availability & Resilience	Strict Quality of Service (QoS) and bandwidth reservation shall be enforced to ensure control loop stability.	To defend against DoS attacks via zero-torque flooding or error frame storms (T486, T487), the network must enforce bandwidth reservation for critical control topics and prioritize them over diagnostic/logging traffic.	T486, T487
	SR-211	Data Confidentiality & Privacy	Control streams containing proprietary tuning parameters (PID) shall be encrypted.	To prevent the theft of "driving feel" calibration data (IP) and leakage of internal controller states via sniffing (T483, T484, T485), control streams carrying PID gains or dynamics parameters must be encrypted, and debug metadata stripped.	T483, T484, T485
DF7. Vehicle Status	SR-212	Authentication & Access Control	Strict source authentication and Access Control Lists (ACLs) shall be enforced for vehicle status publications.	To prevent unauthorized injection of false feedback (e.g., "Wheel Speed: 0") (T488) or masquerading as critical nodes like BMS (T489), the middleware must verify the identity of status publishers and restrict diagnostic access.	T488, T490
	SR-213	Communication Security & Integrity	Status data transmission shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate modification of steering feedback or door status, and replay of old status frames (T489, T491), status packets must include integrity tags (MAC) and freshness counters verified by the receiving controller.	T489, T491
	SR-214	Availability & Resilience	Rate limiting and Error Handling logic shall be implemented to prevent bus saturation from status floods.	To defend against DoS attacks via status update floods or "Error Storms" (T495, T496), the system must enforce rate limits on status messages and implement robust error handling to prevent gateway overload.	T495, T496
	SR-215	Data Confidentiality & Privacy	Diagnostic data (DTC, VIN) and health reports shall be encrypted during transmission and access via ports.	To prevent leakage of vehicle health history and identity via OBD-II tapping or cloud transmission sniffing (T492, T493), diagnostic data streams must be encrypted, and physical ports should have access protection (e.g., Seed/Key exchange).	T492, T493
	SR-216	Data Confidentiality & Privacy	Unnecessary driving history and location metadata shall be minimized in status reports.	To prevent driver profiling (driving style, location) via excessive data collection (T494), status reports sent to the cloud or logs must minimize metadata to what is strictly necessary for maintenance.	T494
DF8. Control Signal	SR-217	Communication Security & Integrity	Control signals shall be protected using hardware-accelerated Message Authentication Codes (MAC) and freshness values (e.g., AUTOSAR SecOC).	To prevent injection of raw frames (T497), replay of braking sequences (T498), and payload bit-flipping (T499), critical control signals must be authenticated using real-time protocols like AUTOSAR SecOC with freshness verification.	T497, T498, T499
	SR-218	Authentication & Access Control	Gateway firewalls and ID whitelisting shall be enforced to separate diagnostic ports from control buses.	To prevent attackers from injecting control IDs directly from the OBD-II port (T497), gateways must enforce strict ID whitelisting and block control messages originating from diagnostic interfaces.	T497
	SR-219	Security Auditing & Monitoring	Intrusion Detection Systems (IDS) shall be implemented to detect bus hogging and error frame storms.	To defend against DoS attacks utilizing priority arbitration (Bus Hogging, T503) or deliberate error frames (Bus Off attack, T504), the system must implement IDS to monitor bus load and error rates, triggering fail-safe modes upon detection.	T503, T504
	SR-220	System Hardening & Platform Security	Calibration protocols (XCP/CCP) shall be disabled in production, and sensitive signals encrypted.	To prevent IP theft (ESP logic) and ECU remapping via calibration signals (T500, T502), debug protocols must be permanently disabled (e.g., Secure Boot fuses) and sensitive signals encrypted where feasible.	T500, T502
	SR-221	System Hardening & Platform Security	Cryptographic keys for signal protection shall be managed securely to prevent side-channel attacks.	To prevent decryption of SecOC frames via weak keys or leakage (T501), keys must be generated with high entropy and stored in Hardware Security Modules (HSM) resistant to side-channel attacks.	T501
DF9. Processed Sensor Data	SR-222	Authentication & Access Control	Middleware shall enforce strict publisher authentication and topic access control (ACLs).	To prevent unauthorized apps from injecting fake object tracks (T505) and masquerading as legitimate detector nodes (T507), the middleware (e.g., DDS/ROS2) must verify the identity of publishers using certificates and restrict topic write access.	T505, T507
	SR-223	Communication Security & Integrity	Processed data streams shall be protected using Message Authentication Codes (MAC) and freshness mechanisms.	To mitigate attribute modification (e.g., velocity change), object classification alteration, and replay attacks (T506, T508), object list payloads must include integrity tags (MAC) and timestamps verified by the fusion/planning module.	T506, T508
	SR-224	AI/Sensor Robustness & Input Validation	Plausibility checks and multi-modal sensor fusion shall be implemented to reject propagated physical spoofing.	To defend against "Ghost Objects" originating from physical sensor spoofing (T509), the consumer module must verify the data's physical plausibility (e.g., laws of physics) and cross-validate with other sensor modalities (Fusion) before acting.	T509
	SR-225	Availability & Resilience	Input validation, rate limiting, and resource quotas shall be enforced to ensure perception availability.	To protect against DoS attacks via object track flooding (T513) or malformed coordinate injection (T514), the system must enforce rate limits, validate data schemas (e.g., check for NaNs), and manage buffer quotas.	T513, T514
	SR-226	Data Confidentiality & Privacy	Processed sensor data streams shall be encrypted to prevent surveillance and IP theft.	To prevent surveillance of surrounding entities via sniffing (T510) and decryption of semantic understanding (T511), data streams containing object lists and classifications must be encrypted.	T510, T511
	SR-227	Data Confidentiality & Privacy	Internal debug metadata (thresholds, pattern IDs) shall be stripped from production data flows.	To prevent attackers from designing adversarial patches using leaked internal logic (T512), debug metadata revealing detection thresholds or pattern matching IDs must be removed from production traffic.	T512
DF10. Planned Trajectory	SR-228	Authentication & Access Control	Middleware shall enforce strict publisher authentication and Access Control Lists (ACLs) for the trajectory topic.	To prevent unauthorized apps from injecting malicious trajectories (e.g., steering into traffic) (T515) or masquerading as the Planner node (T517), the middleware (ROS2/DDS) must enforce strict authentication and allow only the authorized Planner to publish to the /planned_path topic.	T515, T517
	SR-229	Communication Security & Integrity	Trajectory data streams shall be protected using Message Authentication Codes (MAC) and strict freshness checks.	To mitigate modification of speed profiles, shifting of waypoints, and replay of old paths (T516, T518), trajectory payloads must include integrity tags (MAC) and synchronized timestamps verified by the Controller.	T516, T518
	SR-230	Availability & Resilience	Input validation (mathematical sanity checks) and rate limiting shall be enforced to ensure control availability.	To defend against DoS attacks via invalid path flooding (T522) or mathematical errors (e.g., division by zero) (T523), the Controller must validate the mathematical feasibility of paths and enforce rate limits.	T522, T523
	SR-231	Data Confidentiality & Privacy	Trajectory streams containing future movement plans shall be encrypted.	To prevent "Predictive Surveillance" where attackers anticipate the vehicle's movement seconds in advance (T519) and to protect operational strategy (T520), trajectory data must be encrypted during transmission.	T519, T520
	SR-232	Data Confidentiality & Privacy	Internal debug metadata (Optimization Cost, Policy Flags) shall be minimized in production.	To prevent the reverse-engineering of the core path planning algorithm (IP theft) (T521), debug information like cost function values must be stripped from the production trajectory stream.	T521

DF11. Manage Status	SR-233	Authentication & Access Control	Middleware shall enforce strict source authentication and Access Control Lists (ACLs) for mode status topics.	To prevent unauthorized apps from injecting fake mode statuses (e.g., "Auto Mode Engaged") (T524) or masquerading as the Mode Manager (T526), the middleware must verify the identity of publishers and restrict write access to the /mode_status topic.	T524, T526
	SR-234	Communication Security & Integrity	Status messages shall be protected using Message Authentication Codes (MAC) and strict freshness checks.	To mitigate the replay of old "Driver Ready" signals (T525) and modification of "Handover State" (T527), status payloads must include integrity tags (MAC) and synchronized timestamps verified by the receiving module to prevent "Double Disengagement".	T525, T527
	SR-235	Availability & Resilience	Strict Quality of Service (QoS) and prioritization shall be enforced for Takeover Request (ToR) messages.	To defend against DoS attacks via status flooding (T532) or error storms (T533) that delay critical Takeover Requests, the system must prioritize ToR messages over all other status updates and enforce rate limiting on non-critical streams.	T532, T533
	SR-236	Data Confidentiality & Privacy	Status streams containing biometric inferences and internal states shall be encrypted.	To prevent the leakage of sensitive biometric data (Drowsiness/Hands-on) via sniffing (T529) and exposure of internal state machine logic (T530), status streams must be encrypted during transmission.	T529, T530
	SR-237	Authentication & Access Control	Secure session management mechanisms shall be implemented to prevent user profile hijacking.	To prevent attackers from hijacking active user sessions or altering safety settings (T52), the system must use secure session tokens, enforce timeouts, and protect session data integrity.	T52